mechanism determines the overall rate of the reaction.

### 12.1.1 Fundamental concepts in organic reaction mechanism

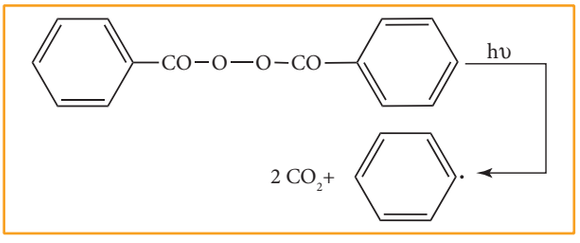
The mechanism is the theoretical pathway which describes the changes occurring in each step during the course of the chemical change. An organic reaction can be understood by following the direction of flow of electrons and the type of intermediate formed during the course of the reaction. The direction of flow of electron is represented by curved arrow. The movement of a pair of electron is represented by a double headed arrow which starts from the negative and ends with the atom to which the electrons needs to be transferred.

### 12.1.2 Fission of a covalent bond

All organic molecules contain covalent bonds which are formed by the mutual sharing of electrons between atoms. These covalent bonds break in two different ways, namely homolytic cleavage (symmetrical splitting) and heterolytic cleavage (unsymmetrical splitting). The cleavage of a bond in the substrate is influenced by the nature of the reagent (attacking agent).

#### Homolytic Cleavage

Homolytic cleavage is the process in which a covalent bond breaks symmetrically in such way that each of the bonded atoms retains one electron. It is denoted by a half headed arrow (fish hook arrow). This type of cleavage occurs under high temperature or in the presence of UV light in a compound containing non polar covalent bond formed between atoms of similar electronegativity. In such molecules, the cleavage of bonds results into free radicals. They are short lived and are highly reactive. The type of reagents that promote homolytic cleavage in substrate are called as free radical initiators. For example Azobisisobutyronitrile (AIBN) and peroxides such as benzoyl peroxide are used as free radical initiators in polymerisation reactions.




As a free radical with an unpaired electron is neutral and unstable, it has a tendency to gain an electron to attain stability. Organic reactions involve homolytic fission of C-C bonds to form alkyl free radicals. The stability of alkyl free radicals is in the following order

$$^{\cdot}C(CH_3)_3 > {}^{\cdot}CH(CH_3)_2 > {}^{\cdot}CH_2CH_3 > {}^{\cdot}CH_3$$

#### Heterolytic Cleavage

Heterolytic cleavage is the process in which a covalent bond breaks unsymmetrically such that one of the bonded atoms retains the bond pair of electrons. It results in the formation of a cation and an anion. Of the two bonded atoms, the most electronegative atom becomes the anion and the other atom becomes the cation. The cleavage is denoted by a curved arrow pointing towards the more electronegative atom.



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